

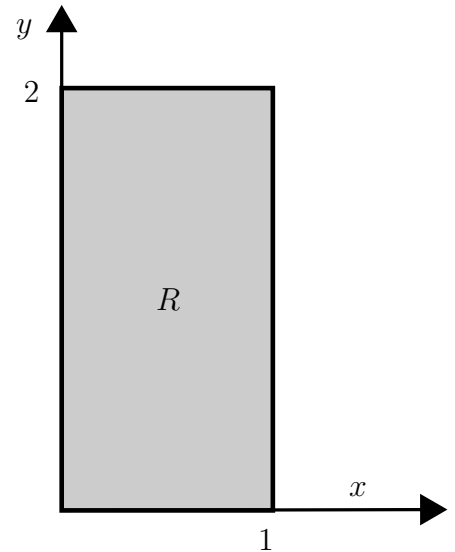
## Multiple Integration

## Solutions to Problem Sheet 4

1. The region of integration  $R$  is sketched below.

Evaluating the integral,

$$\begin{aligned} I &= \int_0^2 \int_0^1 (x + y) \, dx \, dy = \int_0^2 \left[ \frac{x^2}{2} + xy \right]_{x=0}^{x=1} dy, \\ &= \int_0^2 \left( \frac{1}{2} + y \right) dy, \\ &= \left[ \frac{y}{2} + \frac{y^2}{2} \right]_{y=0}^{y=2}, \\ &= 1 + 2 = 3. \end{aligned}$$



2. (a)

$$\begin{aligned} \int_0^1 \int_0^3 z^2 \, dx \, dz &= \int_0^1 [xz^2]_{x=0}^{x=3} dz, \\ &= \int_0^1 3z^2 \, dz, \\ &= [z^3]_{z=0}^{z=1} = 1. \end{aligned}$$

- (b)

$$\begin{aligned} \int_0^4 \int_0^2 x \, dy \, dz &= \int_0^4 [xy]_{y=0}^{y=2} dz, \\ &= \int_0^4 2x \, dz, \\ &= [2xz]_{z=0}^{z=4} = 8x. \end{aligned}$$

(c)

$$\begin{aligned}\int_1^2 \int_2^3 (x^2 y + 1) dx dy &= \int_1^2 \left[ \frac{x^3 y}{3} + x \right]_{x=2}^{x=3} dy, \\ &= \int_1^2 (9y + 3) - \left( \frac{8y}{3} + 2 \right) dy, \\ &= \int_1^2 \frac{19y}{3} + 1 dy, \\ &= \left[ \frac{19y^2}{6} + y \right]_{y=1}^{y=2}, \\ &= \left( \frac{38}{3} + 2 \right) - \left( \frac{19}{6} + 1 \right) = \frac{21}{2} (= 10.5).\end{aligned}$$

(d)

$$\begin{aligned}\int_{-1}^1 \int_2^3 \frac{y}{x} dx dy &= \int_{-1}^1 [y \ln x]_{x=2}^{x=3} dy, \\ &= \int_{-1}^1 y \ln \left( \frac{3}{2} \right) dy, \quad \left[ \ln 3 - \ln 2 = \ln \left( \frac{3}{2} \right) \right] \\ &= \left[ \frac{y^2}{2} \ln \left( \frac{3}{2} \right) \right]_{y=-1}^{y=1}, \\ &= \frac{1}{2} \ln \left( \frac{3}{2} \right) - \frac{1}{2} \ln \left( \frac{3}{2} \right) = 0.\end{aligned}$$

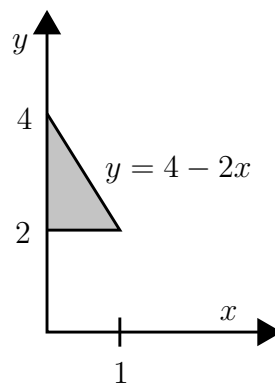
3. (a)

Original integral:

$$\int_0^1 \int_2^{4-2x} dy dx.$$

Reversing the order of integration:

$$\int_2^4 \int_0^{2-y/2} dx dy.$$



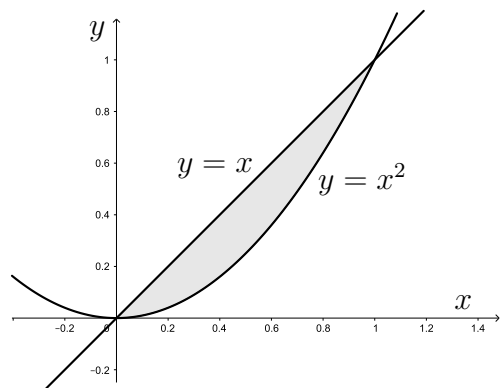
(b)

Original integral:

$$\int_0^1 \int_y^{\sqrt{y}} dx dy.$$

Reversing the order of integration:

$$\int_0^1 \int_{x^2}^x dy dx.$$



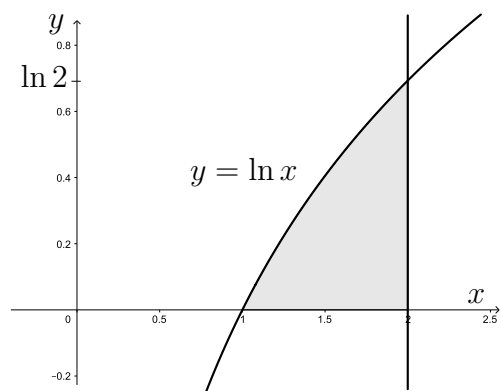
(c)

Original integral:

$$\int_0^{\ln 2} \int_{e^y}^2 dx dy.$$

Reversing the order of integration:

$$\int_1^2 \int_0^{\ln x} dy dx.$$



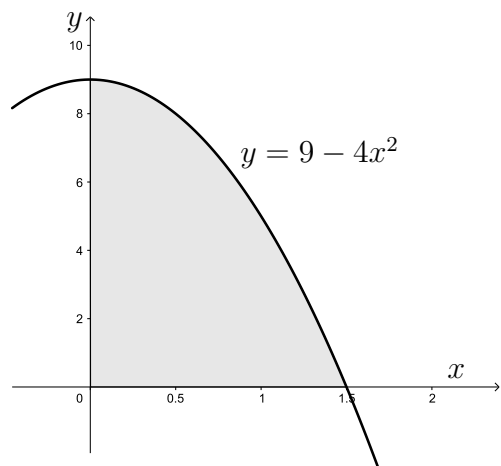
(d)

Original integral:

$$\int_0^{3/2} \int_0^{9-4x^2} 16x dy dx.$$

Reversing the order of integration:

$$\int_0^9 \int_0^{\frac{1}{2}\sqrt{9-y}} 16x dx dy.$$



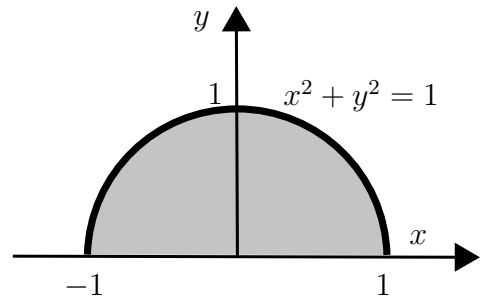
(e)

Original integral:

$$\int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} 3y \, dx \, dy.$$

Reversing the order of integration:

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} 3y \, dy \, dx.$$



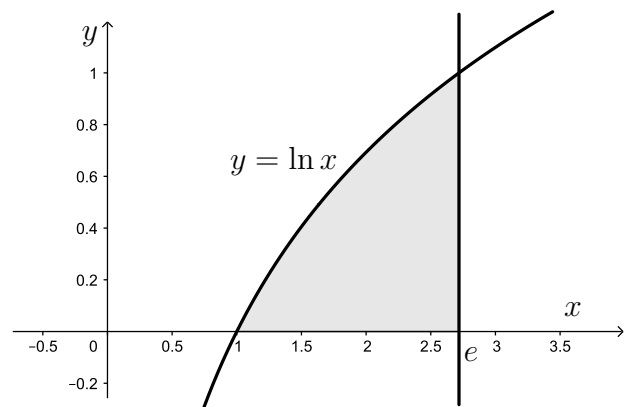
(f)

Original integral:

$$\int_1^e \int_0^{\ln x} xy \, dy \, dx.$$

Reversing the order of integration:

$$\int_0^1 \int_{e^y}^e xy \, dx \, dy.$$



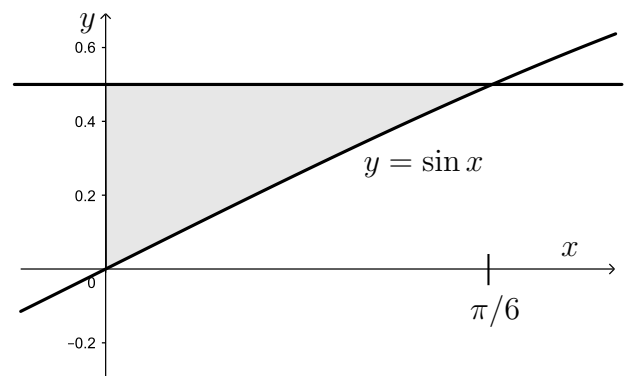
(g)

Original integral:

$$\int_0^{\pi/6} \int_{\sin x}^{1/2} xy^2 \, dy \, dx.$$

Reversing the order of integration:

$$\int_0^{1/2} \int_0^{\sin^{-1} y} xy^2 \, dx \, dy.$$



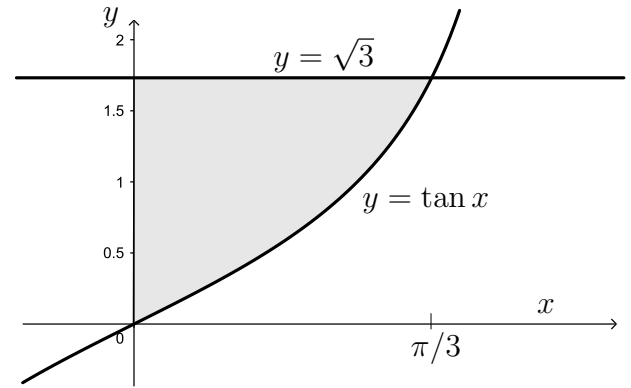
(h)

Original integral:

$$\int_0^{\sqrt{3}} \int_0^{\tan^{-1} y} \sqrt{xy} \, dx \, dy.$$

Reversing the order of integration:

$$\int_0^{\pi/3} \int_{\tan x}^{\sqrt{3}} \sqrt{xy} \, dy \, dx.$$



4. (a)  $\int_1^2 \int_{2y}^4 F(x, y) \, dx \, dy.$

(b)  $\int_2^4 \int_1^{x/2} F(x, y) \, dy \, dx.$

5. (i) (a)  $\int_0^1 \int_0^{y^2} F(x, y) \, dx \, dy.$

(b)  $\int_0^1 \int_{\sqrt{x}}^1 F(x, y) \, dy \, dx.$

(ii) (a)  $\int_0^1 \int_{-\sqrt{1-y}}^{\sqrt{1-y}} F(x, y) \, dx \, dy.$

(b)  $\int_{-1}^1 \int_0^{1-x^2} F(x, y) \, dy \, dx.$

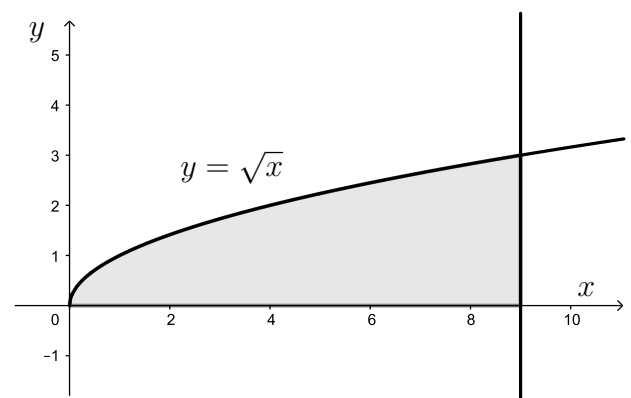
6. (a)

Inner integral with respect to  $x$ :

$$\int_0^3 \int_{y^2}^9 F(x, y) \, dx \, dy.$$

Inner integral with respect to  $y$ :

$$\int_0^9 \int_0^{\sqrt{x}} F(x, y) \, dy \, dx.$$



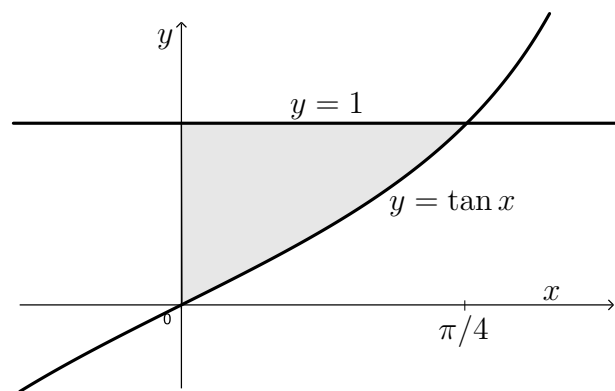
(b)

Inner integral with respect to  $x$ :

$$\int_0^1 \int_0^{\tan^{-1} x} F(x, y) \, dx \, dy.$$

Inner integral with respect to  $y$ :

$$\int_0^{\pi/4} \int_{\tan x}^1 F(x, y) \, dy \, dx.$$



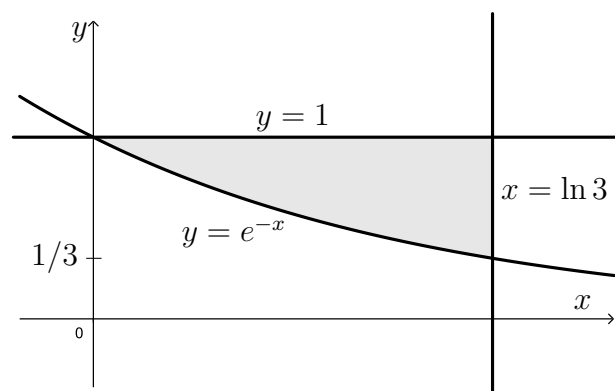
(c)

Inner integral with respect to  $x$ :

$$\int_{1/3}^1 \int_{-\ln y}^{\ln 3} F(x, y) \, dx \, dy.$$

Inner integral with respect to  $y$ :

$$\int_0^{\ln 3} \int_{e^{-x}}^1 F(x, y) \, dy \, dx.$$



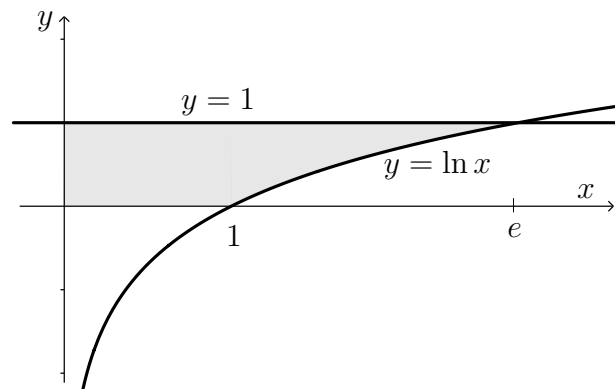
(d)

Inner integral with respect to  $x$ :

$$\int_0^1 \int_0^{e^y} F(x, y) \, dx \, dy.$$

Inner integral with respect to  $y$ :

$$\int_0^1 \int_0^1 F(x, y) \, dy \, dx + \int_1^e \int_{\ln x}^1 F(x, y) \, dy \, dx.$$



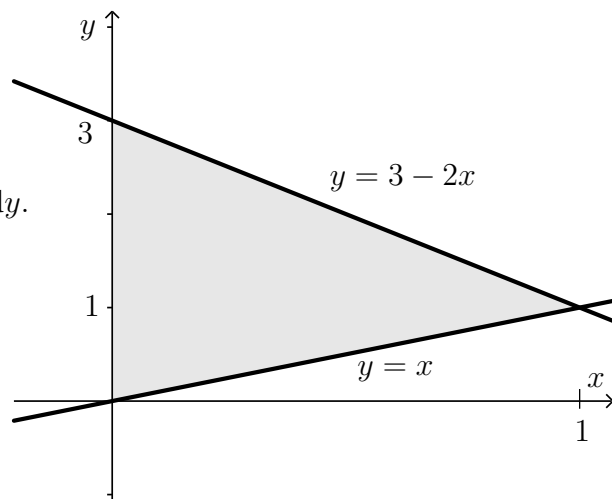
(e)

Inner integral with respect to  $x$ :

$$\int_0^1 \int_0^y F(x, y) dx dy + \int_1^3 \int_0^{(3-y)/2} F(x, y) dx dy.$$

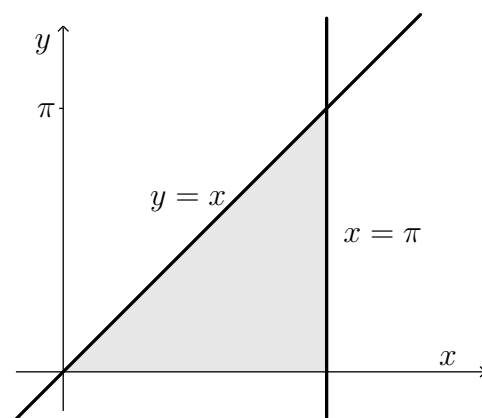
Inner integral with respect to  $y$ :

$$\int_0^1 \int_x^{3-2x} F(x, y) dy dx.$$



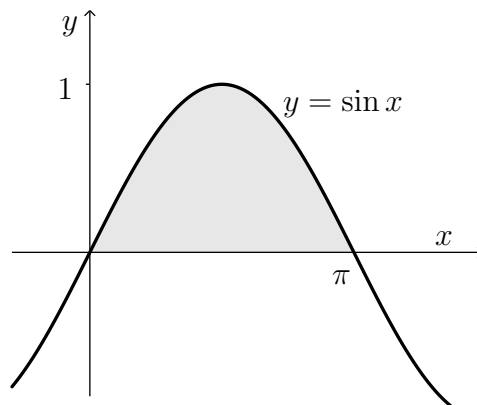
7. (a) Evaluating the integral,

$$\begin{aligned} \int_0^\pi \int_0^x x \sin y dy dx &= \int_0^\pi [-x \cos y]_{y=0}^{y=x} dx, \\ &= \int_0^\pi x - x \cos x dx, \\ \text{(int. by parts)} \quad &= \left[ \frac{x^2}{2} - (x \sin x + \cos x) \right]_{x=0}^{x=\pi}, \\ &= \left( \frac{\pi^2}{2} + 1 \right) - (-1), \\ &= \frac{\pi^2}{2} + 2. \end{aligned}$$



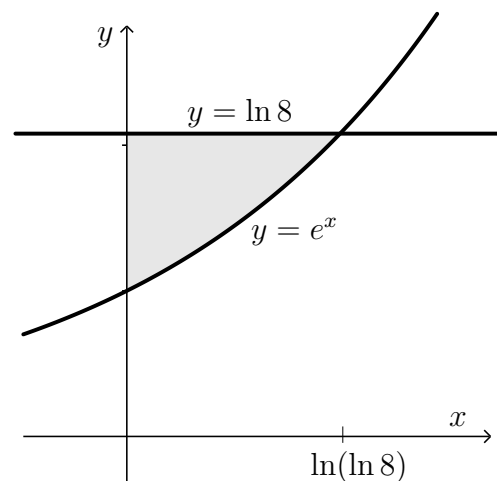
(b) Evaluating the integral,

$$\begin{aligned} \int_0^\pi \int_0^{\sin x} y dy dx &= \int_0^\pi \left[ \frac{y^2}{2} \right]_{y=0}^{y=\sin x} dx, \\ &= \frac{1}{2} \int_0^\pi \sin^2 x dx, \\ &= \frac{1}{2} \int_0^\pi \frac{1}{2} (1 - \cos 2x) dx, \\ &= \frac{1}{4} \left[ x - \frac{1}{2} \sin 2x \right]_{x=0}^{x=\pi}, \\ &= \frac{\pi}{4}. \end{aligned}$$



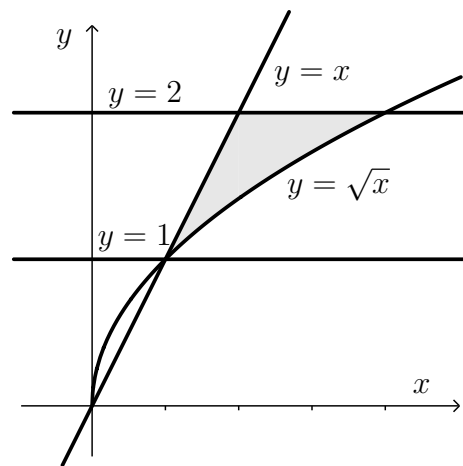
(c) Evaluating the integral,

$$\begin{aligned}
 \int_1^{\ln 8} \int_0^{\ln y} e^x e^y dx dy &= \int_1^{\ln 8} [e^x e^y]_{x=1}^{x=\ln y} dy, \\
 &= \int_1^{\ln 8} ye^y - e^y dy, \\
 (\text{int. by parts}) &= [ye^y - 2e^y]_{y=1}^{y=\ln 8}, \\
 &= (8 \ln 8 - 16) - (e - 2e), \\
 &= 8 \ln 8 - 16 + e.
 \end{aligned}$$



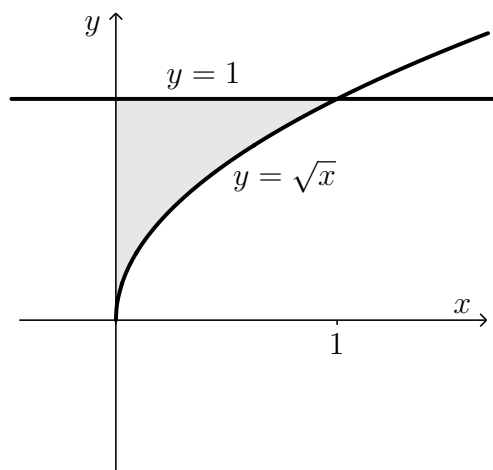
(d) Evaluating the integral,

$$\begin{aligned}
 \int_1^2 \int_y^{y^2} dx dy &= \int_1^2 [x]_{x=y}^{x=y^2} dy, \\
 &= \int_1^2 y^2 - y dy, \\
 &= \left[ \frac{y^3}{3} - \frac{y^2}{2} \right]_{y=1}^{y=2}, \\
 &= \left( \frac{8}{3} - 2 \right) - \left( \frac{1}{3} - \frac{1}{2} \right), \\
 &= \frac{5}{6}.
 \end{aligned}$$



(e) Evaluating the integral,

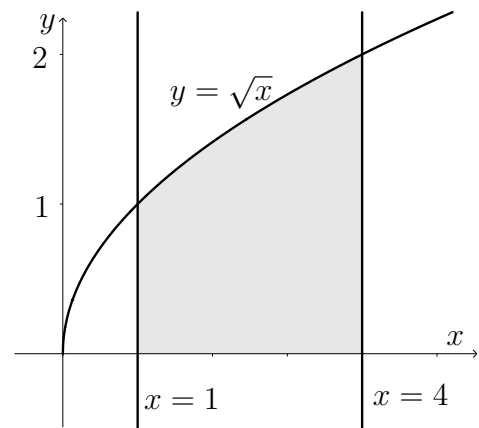
$$\begin{aligned}
 \int_0^1 \int_0^{y^2} 3y^3 e^{xy} dx dy &= \int_0^1 [3y^2 e^{xy}]_{x=0}^{x=y^2} dy, \\
 &= \int_0^1 3y^2 e^{y^3} - 3y^2 dy, \\
 &= [e^{y^3} - y^3]_{y=0}^{y=1}, \\
 &= (e - 1) - 1 = e - 2.
 \end{aligned}$$



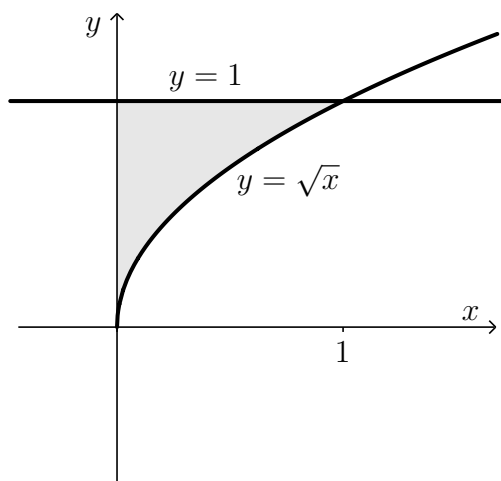


(f) Evaluating the integral,

$$\begin{aligned}
 \int_1^4 \int_0^{\sqrt{x}} \frac{3}{2} e^{y/\sqrt{x}} dy dx &= \int_1^4 \left[ \frac{3\sqrt{x}}{2} e^{y/\sqrt{x}} \right]_{y=0}^{y=\sqrt{x}} dx, \\
 &= \int_1^4 \frac{3ex^{1/2}}{2} - \frac{3x^{1/2}}{2} dx, \\
 &= [ex^{3/2} - x^{3/2}]_{x=1}^{x=4}, \\
 &= (e-1)(8-1), \\
 &= 7(e-1).
 \end{aligned}$$



8. The region  $A$  is sketched below (and is in fact the same as in question 7(e) above).



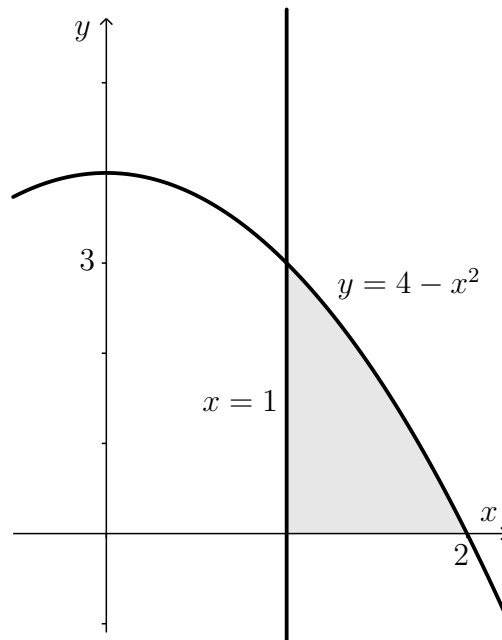
(a) Inner integral with respect to  $x$ :

$$\begin{aligned}
 \int_0^1 \int_0^{y^2} 2xy - x^2 dx dy &= \int_0^1 \left[ x^2y - \frac{x^3}{3} \right]_{x=0}^{x=y^2} dy, \\
 &= \int_0^1 y^5 - \frac{y^6}{3} dy, \\
 &= \left[ \frac{y^6}{6} - \frac{y^7}{21} \right]_{y=0}^{y=1}, \\
 &= \frac{1}{6} - \frac{1}{21} = \frac{5}{42}.
 \end{aligned}$$

(b) Inner integral with respect to  $y$ :

$$\begin{aligned}
 \int_0^1 \int_{\sqrt{x}}^1 2xy - x^2 \, dy \, dx &= \int_0^1 [xy^2 - x^2y]_{y=\sqrt{x}}^{y=1} \, dx, \\
 &= \int_0^1 x - x^2 - (x^2 - x^{5/2}) \, dx, \\
 &= \int_0^1 x - 2x^2 + x^{5/2} \, dx, \\
 &= \left[ \frac{x^2}{2} - \frac{2x^3}{3} + \frac{2x^{7/2}}{7} \right]_{x=0}^{x=1}, \\
 &= \frac{1}{2} - \frac{2}{3} + \frac{2}{7} = \frac{5}{42}.
 \end{aligned}$$

9. The region of integration is sketched below.



Evaluating the integral:

$$\begin{aligned}
 \int_0^3 \int_1^{\sqrt{4-y}} x + y \, dx \, dy &= \int_0^3 \left[ \frac{x^2}{2} + xy \right]_{x=1}^{x=\sqrt{4-y}} \, dy, \\
 &= \int_0^3 2 - \frac{y}{2} + y\sqrt{4-y} - \left( \frac{1}{2} + y \right) \, dy, \\
 &= \int_0^3 \frac{3}{2} - \frac{3y}{2} + y(4-y)^{1/2} \, dy, \\
 \text{(int. by parts, see below)} &= \left[ \frac{3y}{2} - \frac{3y^2}{4} - \frac{2y}{3}(4-y)^{3/2} - \frac{4}{15}(4-y)^{5/2} \right]_{y=0}^{y=3}, \\
 &= \left( \frac{9}{2} - \frac{27}{4} - 2 - \frac{4}{15} \right) + \frac{128}{15}, \\
 &= \frac{241}{60}
 \end{aligned}$$

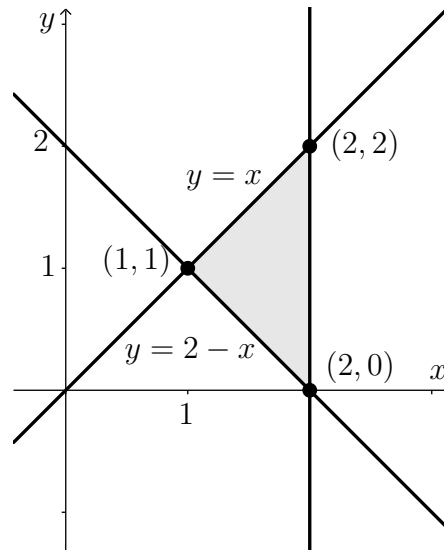
Details of the integration by parts for  $\int y(4-y)^{1/2} dy$ . Let  $u = y$ ,  $v' = (4-y)^{1/2}$ . Then  $u' = 1$ ,  $v = -\frac{2}{3}(4-y)^{3/2}$ .

$$\begin{aligned}\int y(4-y)^{1/2} dy &= -\frac{2}{3}y(4-y)^{3/2} + \frac{2}{3} \int (4-y)^{3/2} dy, \\ &= -\frac{2}{3}y(4-y)^{3/2} - \frac{4}{15}(4-y)^{5/2} + c.\end{aligned}$$

Reversing the order of integration and evaluating again:

$$\begin{aligned}\int_1^2 \int_0^{4-x^2} x+y \, dy \, dx &= \int_1^2 \left[ xy + \frac{y^2}{2} \right]_{y=0}^{y=4-x^2} dx, \\ &= 4x - x^3 + \frac{(4-x^2)^2}{2} dx, \\ &= 4x - x^3 + 8 - 4x^2 + \frac{x^4}{2} dx, \\ &= \left[ 2x^2 - \frac{x^4}{4} + 8x - \frac{4x^3}{3} + \frac{x^5}{10} \right]_{x=1}^{x=2}, \\ &= \left( 8 - 4 + 16 - \frac{32}{3} + \frac{32}{10} \right) - \left( 2 - \frac{1}{4} + 8 - \frac{4}{3} + \frac{1}{10} \right), \\ &= \frac{241}{60}.\end{aligned}$$

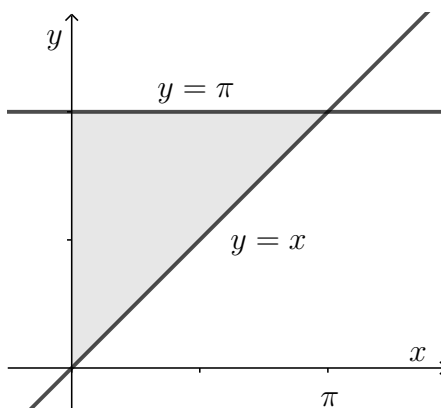
10. The region  $A$  is sketched below.



The simplest way to evaluate the integral is for the inner integral to be with respect to  $y$ .

$$\begin{aligned}
 \int_1^2 \int_{2-x}^x 5x^2 + 2y^2 \, dy \, dx &= \int_1^2 \left[ 5x^2 y + \frac{2y^3}{3} \right]_{y=2-x}^{y=x} dx, \\
 &= \int_1^2 5x^3 + \frac{2x^3}{3} - \left( 5x^2(2-x) + \frac{2(2-x)^3}{3} \right) dx, \\
 &= \int_1^2 \frac{17x^3}{3} - \left( 10x^2 - 5x^3 + \frac{2}{3}(8 - 12x + 6x^2 - x^3) \right) dx, \\
 &= \int_1^2 \frac{34x^3}{3} - 14x^2 + 8x - \frac{16}{3} dx, \\
 &= \left[ \frac{17x^4}{6} - \frac{14x^3}{3} + 4x^2 - \frac{16x}{3} \right]_{x=1}^{x=2}, \\
 &= \left( \frac{136}{3} - \frac{112}{3} + 16 - \frac{32}{3} \right) - \left( \frac{17}{6} - \frac{14}{3} + 4 - \frac{16}{3} \right) = \frac{33}{2}.
 \end{aligned}$$

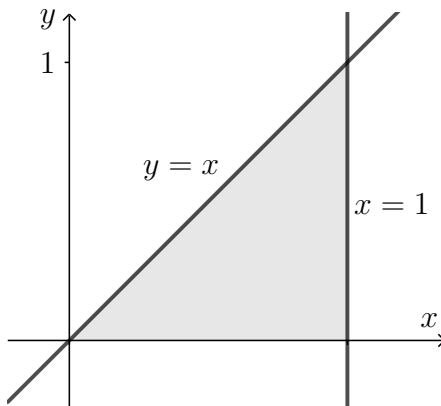
11. (a) The region is sketched below.



Reversing the order of integration and evaluating:

$$\begin{aligned}
 \int_0^\pi \int_x^\pi \frac{\sin y}{y} \, dy \, dx &= \int_0^\pi \int_0^y \frac{\sin y}{y} \, dx \, dy, \\
 &= \int_0^\pi \left[ \frac{x \sin y}{y} \right]_{x=0}^{x=y} dy, \\
 &= \int_0^\pi \sin y \, dy, \\
 &= [-\cos y]_{x=0}^{x=\pi} = 1 - (-1) = 2.
 \end{aligned}$$

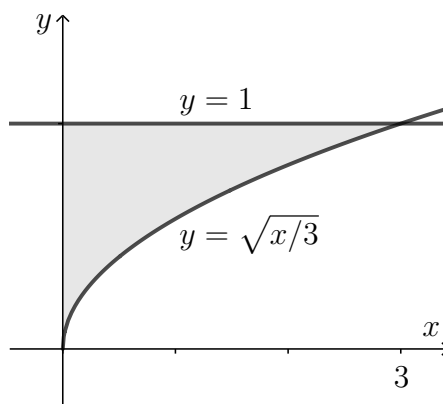
(b) The region is sketched below.



Reversing the order of integration and evaluating:

$$\begin{aligned}
 \int_0^1 \int_y^1 x^2 e^{xy} \, dx \, dy &= \int_0^1 \int_0^x x^2 e^{xy} \, dy \, dx, \\
 &= \int_0^1 [xe^{xy}]_{y=0}^{y=x} \, dx, \\
 &= \int_0^1 xe^{x^2} - x \, dx, \\
 &= \left[ \frac{1}{2}e^{x^2} - \frac{x^2}{2} \right]_{x=0}^{x=1} = \frac{e}{2} - \frac{1}{2} - \frac{1}{2} = \frac{e}{2} - 1.
 \end{aligned}$$

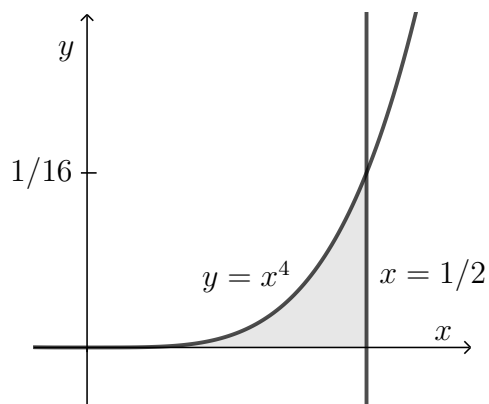
(c) The region is sketched below.



Reversing the order of integration and evaluating:

$$\begin{aligned}
 \int_0^3 \int_{\sqrt{x/3}}^1 e^{y^3} \, dy \, dx &= \int_0^1 \int_0^{3y^2} e^{y^3} \, dx \, dy, \\
 &= \int_0^1 [xe^{y^3}]_{x=0}^{x=3y^2} \, dy, \\
 &= \int_0^1 3y^2 e^{y^3} \, dy = [e^{y^3}]_{y=0}^{y=1} = e - 1.
 \end{aligned}$$

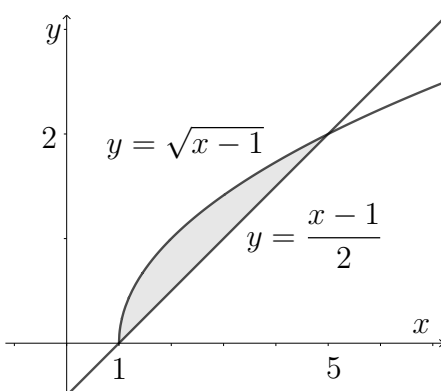
(d) The region is sketched below.



Reversing the order of integration and evaluating:

$$\begin{aligned}
 \int_0^{1/16} \int_{y^{1/4}}^{1/2} \cos(16\pi x^5) \, dx \, dy &= \int_0^{1/2} \int_0^{x^4} \cos(16\pi x^5) \, dy \, dx, \\
 &= \int_0^{1/2} [y \cos(16\pi x^5)]_{y=0}^{y=x^4} \, dx, \\
 &= \int_0^{1/2} x^4 \cos(16\pi x^5) \, dx, \\
 &= \frac{1}{80\pi} [\sin(16\pi x^5)]_{x=0}^{x=1/2}, \\
 &= \frac{1}{80\pi} (1 - 0) = \frac{1}{80\pi}.
 \end{aligned}$$

12. The region  $A$  is sketched below.



Evaluating the integral:

$$\begin{aligned}
 \int_0^2 \int_{y^2+1}^{2y+1} (x+2y)^{-1/2} \, dx \, dy &= \int_0^2 [2(x+2y)^{1/2}]_{x=y^2+1}^{x=2y+1} \, dy, \\
 &= 2 \int_0^2 (4y+1)^{1/2} - (y^2+2y+1)^{1/2} \, dy, \\
 &= 2 \int_0^2 (4y+1)^{1/2} - ((y+1)^2)^{1/2} \, dy, \\
 &= 2 \int_0^2 (4y+1)^{1/2} - (y+1) \, dy, \\
 &= 2 \left[ \frac{1}{6} (4y+1)^{3/2} - \frac{y^2}{2} - y \right]_{y=0}^{y=2}, \\
 &= 2 \left( \frac{9}{2} - 2 - 2 - \frac{1}{6} \right) = \frac{2}{3}.
 \end{aligned}$$